Easy point wise

First Fit Memory Allocation

- It is a memory allocation technique in which the process is placed in the **first free memory block** that is large enough.
- It is simple and fast.
- It may cause memory wastage due to fragmentation.

Worst Fit Memory Allocation

- It is a memory allocation technique in which the process is placed in the **largest available memory block**.
- It leaves larger free spaces after allocation.
- It may reduce the chance of small unused spaces.

Logical address and physical address

Logical Address and Physical Address in Operating System

Logical Address

- A **logical address** is the address generated by the CPU while a program is running.
- It is also called a **virtual address**.
- The user/program can see this address.
- It is converted into a physical address by the memory management unit (MMU).

Physical Address

- A **physical address** is the actual address location in the main memory (RAM).
- It is the address used by the hardware to access data.
- It is not directly visible to the user.

Preemptive Scheduling

- In this scheduling method, the operating system can **interrupt a running process** and give the CPU to another process.
- A process can be stopped before completing its task.
- It improves response time.
- It is used in time-sharing systems.

Examples: Round Robin, Shortest Remaining Time First (SRTF), Priority Scheduling.

Non-Preemptive Scheduling

- In this scheduling method, a process keeps the CPU until it **completes its execution** or enters a waiting state.
- The operating system cannot forcefully stop the running process.
- It is simple and has less overhead.

Examples: First Come ... Serve (FCFS), Shortest Job First (SJF).

IPC (Inter Process Communication)

Definition:
IPC is a method used in an operating system that allows **different processes to communicate and share information** with each other.

Types of IPC:

1. **Shared Memory**

- Processes share the same memory area.
- It allows fast communication.

2. **Message Passing**

- Processes send and receive messages to communicate.

Advantages:

- Helps processes work together.
- Allows data sharing between processes.
- Improves system performance.

A semaphore is a synchronization tool used by the operating system to control access to shared resources by multiple processes.

Points:

- It helps prevent conflicts when multiple processes access the same resource.
- It allows only a limited number of processes to use a resource at a time.
- It is used to solve the critical section problem.

Types of Semaphore:

1. **Binary Semaphore**
   - Has only two values: 0 and 1.
   - Used for mutual exclusion.
2. **Counting Semaphore**
   - Can have multiple values.
   - Used to control access to multiple resources.

Example:
A semaphore can control access to a printer so that only one process uses it at a time.

2 marks

Thrashing

- Thrashing is a condition where the operating system spends more time **swapping pages** than executing processes.
- It occurs due to insufficient memory and causes slow system performance.

Virtual Memory

- Virtual memory is a memory management technique that uses **secondary storage as an extension of RAM**.
- It allows large programs to run even when the physical memory is limited.

I think you mean Wait-for Graph in Operating System.

Wait-for Graph

Definition:
A **Wait-for Graph** is a graph used to detect **deadlock** in an operating system. It shows which process is waiting for another process to release a resource.

Diagram Example:

```
P1  ———————→  P2
       ↑          |
       |          ↓
       P3  ←————  P4
```

Explanation:

- **P1 is waiting for P2**
- **P2 is waiting for P4**
- **P4 is waiting for P3**
- **P3 is waiting for P1**

This creates a **cycle**, so a **deadlock occurs**.

Points:

Explain any 2 directory structure

A **directory structure** is the way files and folders are organized on a computer system. Two common types are:

1. Single-Level Directory Structure

- All files are stored in one single directory (folder).
- It is simple and easy to manage for small systems.
- File names must be unique because there is only one directory.

Example:

```
Root Directory
│
├── file1.txt
├── file2.txt
├── image.jpg
└── program.exe
```

Ad

- Simple design

```
Root Directory
│
├── file1.txt
├── file2.txt
├── image.jpg
└── program.exe
```

Advantages:

- Simple design
- Easy to implement

Disadvantages:

- Difficult to manage many files
- Name conflicts can occur

2. Tree Directory Structure

- Files are organized in a hierarchy of directories and subdirectories.
- It is the most commonly used structure in modern operating systems.

Example:

```
Root
│
├── Documents
│   ├── notes.txt
│   └── report.doc
│
```

...ost commonly used struct... modern operating systems.

Example:

```
Root
│
├── Documents
│   ├── notes.txt
│   └── report.doc
│
├── Pictures
│   └── photo.jpg
│
└── Music
    └── song.mp3
```

Advantages:

- Better organization of files
- Easier searching and management
- Allows users to group related files

Disadvantages:

- More complex than a single-level directory structure
- Requires more management

These are two basic directory structures used in operating systems.

- **SCAN (Disk Scheduling):**
  - Also called **Elevator algorithm**.
  - Disk arm moves in one direction servicing requests, then reverses direction.
- **C-SCAN (Circular SCAN):**
  - Disk arm moves in only **one direction** servicing requests.
  - After reaching the end, it jumps back to the beginning without servicing.
- **LOOK:**
  - Similar to SCAN.
  - The disk arm goes only up to the **last pending request** in that direction, not the end of the disk.
- **C-LOOK:**
  - Similar to C-SCAN.
  - Moves in one direction up to the last request, then returns to the first request.
- **FIFO (First In First Out):**
  - The first process, [illegible] that enters is removed first.

- Similar to C-SCAN.
- Moves in one direction up to the last request, then returns to the first request.

- **FIFO (First In First Out):**
  - The first process/page that enters is removed first.
  - Uses a simple queue method.
- **STF / SSTF (Shortest Seek Time First):**
  - Selects the disk request closest to the current head position.
  - Reduces seek time compared to FIFO.
- **LRU (Least Recently Used):**
  - Removes the page that has not been used for the longest time.
  - Used in page replacement algorithms.
- **Optimal Page Replacement:**
  - Replaces the page that will not be used for the longest time in the future.
  - Gives the minimum number of page faults.
- **SCAN (again):**
  - Services disk requests in a fixed direction and then changes direction like an elevator.